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MILITARY SPECIFICATION

BEACON, RADAR AN/APN-154(V)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The equipment covered by this specification shall receive interrogations from X-band radars and transmit a reply to the interrogation on another frequency within the X-band. It shall also transmit X-band pulses when triggered by the associated equipment (see 6.12) in the "All-Weather Carrier Landing System (ACLS)" mode of operation.

1.2 Classification. A radar beacon AN/APN-154(V) shall consist of the following items in the quantity indicated:

Item	Quantity
Receiver-Transmitter, Radar	1
Control	1
Duplexer	1
Cable Assembly, Radio Frequency	2

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, Systems Requirements Department, Code SR3, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Receiver-transmitter, radar. The receiver-transmitter shall be available in the following configurations. The particular configuration is to be specified in the contract or purchase order (see 6.11).

Type designation	Brief description	Applicable paragraph
RT-681/APN-154(V)	High band receiver - flange mount (see figure 1)	3.5.1 table I
RT-762/APN-154(V)	Low band receiver - side mount (see figure 2)	3.5.1 table I
RT-763/APN-154(V) 1/	High band receiver - side mount (see figure 2)	3.5.1 table I
RT-764/APN-154(V)	Low band receiver - flange mount (see figure 1)	3.5.1 table I
RT-681A/APN-154(V)	High band receiver - flange mount and elapsed time indicator (see figure 3)	3.5.1 table I
RT-762A/APN-154(V)	Low band receiver - side mount with elapsed time indicator (see figure 4)	3.5.1 table I
RT-763A/APN-154(V) 1/	High band receiver - side mount and elapsed time indicator (see figure 4)	3.5.1 table I
RT-764A/APN-154(V)	Low band receiver - flange mount with elapsed time indicator (see figure 3)	3.5.1 table I
RT-763B/APN-154(V) 1/	High band receiver - side mount and elapsed time indicator (see figure 4)	3.5.1
RT-763C/APN-154(V) 1/	High band receiver - side mount and elapsed time indicator (see figure 4)	3.5.1
RT-763D/APN-154(V) 1/	High band receiver - side mount and elapsed time indicator (see figure 4 and 1.3.1)	3.5.1

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RT-763E/APN-154(V) 1/	High band receiver - side mount and elapsed time indicator (see figure 4 and 1.3.2)	3.5.1 table I

1/ This specification revision is not to be used for future RT-763/APN-154 designs.

1.3.1 ACLS operation. The RT-763D/APN-154(V) and the RT-763/APN-154(V) configurations incorporate external trigger capability for ACLS operation.

1.3.2 Solid-state oscillator. The RT-763E/APN-154(V) configuration contains a solid-state oscillator.

1.4 Control. The control shall be available in the following configurations, which shall be specified in the contract or purchase order.

Type designation	Brief description	Applicable paragraph
C-4419/APN-154(V)	Control without elapsed time indicator	3.5.2
C-4419/APN-154(V)	Control with elapsed time indicator	3.5.5
C-4419B/APN-154(V)	Control with ACLS test light and switch	3.5.6

1.5. Duplexer. The duplexer shall be available in the following configuration.

Type designation	Brief description	Applicable paragraph
CU-1104/APN-154(V)	Duplexer - circulator 4-port	3.5.3

1.6 Cable assembly. The radio frequency (rf) cable assemblies shall be available in the following configurations.

Type designation	Brief description	Applicable paragraph
CG-2574/APN-154(V)	RF cable - 6 inches long	3.5.4
CG-2574A/APN-154(V)	RF cable - 8 inches long	3.5.4

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2. APPLICABLE DOCUMENTS

* 2.1 Government documents

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-W-5088	-	Wiring, Aerospace Vehicle
MIL-C-6781	-	Control Panel, Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	-	Panels, Information, Integrally Illuminated
MIL-E-17555	-	Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of
MIL-T-18303	-	Test Procedures Reproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	-	Nomenclature and Identification for Aeronautical Systems Including Joint Electronics Type Designated Systems and Associated Support Systems

STANDARDS

MILITARY

MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference

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STANDARDS

MILITARY (continued)

MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	-	Aircraft Electric Power Characteristics
MIL-STD-781	-	Reliability Testing for Engineering Development, Qualification,a nd Production
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-5400	-	Electronic Equipment, Airborne, General Specification For

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.3), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 Parts and materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In doing so, the following shall govern:

a. Parts and materials requirements shall conform to MIL-STD-5400.

b. Nonrepairable subassemblies, as outlined in MIL-STD-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the acquiring activity. Nonrepairable subassemblies must be reliable (see 6.8).

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c. Repairable subassembly design incorporated in previous procurements shall be changed to nonrepairable only upon approval of the acquiring activity.

3.2.1 Nonstandard parts and materials approval. Approval for the use of nonstandard parts and materials shall be obtained as specified in MIL-STD-5400.

3.3 Design and construction. Unless otherwise specified herein, the equipment shall conform to all the applicable requirements of MIL-STD-5400 for design, construction, and workmanship.

3.3.1 Subminiature construction. Semi-conductor devices shall be used throughout the equipment to the greatest extent possible and the use of vacuum tubes shall be kept to a minimum. The modulator shall be of the semi-conductor type. The use of the latest type silicon junction transistors shall be considered for this equipment.

3.3.1.1 Subassemblies. The design, layout, and assembly of the subassemblies and their component parts shall be such as to facilitate quantity production and to result in minimum size and weight. Microminiature construction techniques shall be utilized to the fullest extent possible consistent with dependability, maintenance requirements, reliability, and the performance of this specification. The design shall include, when possible, circuits that have been or can be reproduced by mechanized or semi-mechanized production facilities consistent with the state of the art. The following types of construction are considered to be classed in the mechanized or semi-mechanized category.

a. Subassemblies using printed circuits, upon which the parts are printed or mechanically placed and soldered.

b. Construction in which integrated circuit techniques are utilized (process whereby circuit components such as transistors, diodes, resistors, capacitors, etc., are formed by use of semi-conductor fabrication techniques, either separately or in combination) such that entire circuit stages are included in a single replaceable package. The contractor shall inform the acquiring activity of those circuits he proposes to make as mechanized assemblies and shall obtain approval therefor.

3.3.2 Total weight. The total weight of the equipment shall be the minimum consistent with good design and will vary with the particular configuration specified in the contract. No configuration shall exceed 6.3 pounds.

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* 3.3.3 Interference control. The generation of radio frequency interference by the equipment and the vulnerability of the equipment to radio frequency interference shall be controlled within the limits specified in MIL-STD-461 and MIL-STD-462. The limits shall be met when the equipment is subjected to the procedure in MIL-STD-462, except that the rf output of any transmitter key up or receiver shall not exceed 50 decibels (db) above 1 microvolt for CW or 70 db above 1 microvolt per megacycle for pulse CD interference at any frequency between 0.15 and 10,000 megacycles.

3.3.1.1 Receiver front-end rejection. Front-end rejection of the receiver shall be equal to or greater than the limit shown on figure 5, except that at the image frequency, the rejection shall be at least 40 db.

3.3.4 Operating life

3.3.4.1 Operational stability. The equipment shall be capable of operating with optimal performance for a total of 150 hours continuously, or intermittently, without the necessity of readjustment of any controls which are inaccessible during flight.

3.3.4.2 Total operating life. The equipment shall have a total operating life of not less than 6000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval, and the life of such parts, shall be specified by the manufacturer.

3.3.4.3 Reliability in mean time between failures. The equipment shall have not less than 500 hours of mean (operating) time between failure when tested and accepted as outlined under the requirements of 4.5.3.

3.3.5 Control panels. All rack or console mounted control panels shall conform to the applicable requirements of MIL-C-6781. The configuration of all control panels must be approved by the acquiring activity prior to first article testing.

3.3.6 Interchangeability. The equipment shall meet the interchangeability requirements specified in MIL-STD-5400.

3.3.7 Maintenance provisions and field testing. Provisions for maintenance shall be as specified in MIL-STD-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.8 Nomenclature and nameplates. Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.3.9 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

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Temperature	Room ambient (25° ±5° C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90 percent relative humidity
Input Power Voltage	28 ±1 Vdc

3.3.10 Primary input power requirements. The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

a. DC power 28V Category "B" 1.30 amps.

3.3.10.1 Under voltage protection. The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.3.11 Service conditions. Except as modified herein, the equipment shall operate satisfactorily under any of the environmental service conditions, or reasonable combination of these conditions, as specified in MIL-STD-5400 for class 2 equipment.

3.3.11.1 Altitude. The maximum altitude shall be increased to 100,000 feet.

3.3.11.2 Vibration. The equipment shall operate satisfactorily when subjected to the vibration requirements of MIL-STD-5400, Curve IV, with the frequency limit extended to 2000 Hz. Vibrations isolators shall not be used.

3.3.11.3 Shock. When normally mounted, the transponder set shall be capable of withstanding, without damage, shock impact up to 100g in any direction.

3.3.11.4 Acceleration. When normally mounted, the transponder set shall be capable of withstanding, without damage, constant acceleration of 100g applied along any axis.

3.3.11.5 Temperature. The equipment shall operate satisfactorily under the temperature vs. altitude conditions of MIL-STD-5400 for continuous operation except that column II of table II shall not apply. It should be noted that the temperature specified herein are heat sink temperatures measured at the mounting flange of the receiver-transmitter.

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3.3.12 Warm-up time. The time required for the equipment to warm up prior to operations shall be kept to a minimum and shall not exceed five minutes under standard conditions and fifteen minutes at extreme service conditions.

3.4 Performance

3.4.1 General. Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under standard conditions will be specified.

3.4.2 Characteristics at X-band. The equipment shall meet the performance standards as follows.

3.4.2.1 Receiver. The center frequency of the receiver pass band shall be adjustable to any frequency from 8500 to 9600 MHz by the installation of one or two interchangeable local oscillators. Each local oscillator shall be adjustable over a frequency range of not less than 500 MHz.

3.4.2.1.1 High band receiver. RT-681/APN-154(V), RT-681A/APN-1254(V), RT-763-APN/154(V), and RT-763A/APN-154(V) configurations shall tune from 9000 to 9600 MHz.

3.4.2.1.2 Low band receiver. RT-762/APN-154(V), Rt-762A/APN-154(V), RT-764/APN-154(V), and RT-764A/APN-154(V) configurations shall tune from 8500 to 9100 MHz.

3.4.2.1.3 High band receiver. RT-763B/APN-154(V), RT-763C/APN-154(V), RT-763D/APN-154(V), and RT-763E/APN-154(V) shall tune from 9000 to 9600 MHz. Unless otherwise specified in the contract, the receiver shall be adjusted to 9310 ±2 MHz when shipped.

3.4.2.2 Transmitter. The transmitter frequency shall be adjustable to any frequency from 8800 to 9500 MHz. Unless otherwise specified in the contract, the receiver shall be adjusted to 9310 ±2 MHz when shipped.

3.4.2.4 Receiving and transmitting operation. The receiver-transmitter shall have separate receiver and transmitter antenna terminals such that it may be used with separate receiving and transmitting antennas. The transmitting antenna shall be separated from the receiving antenna so that at least 40 db of isolation shall occur.

3.4.2.4.1 Delay. Delay is defined as the time delay from the leading edge of the last pulse of the interrogating code group to the landing edge of the transmitted pulse.

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3.4.2.4.2 Fixed delay. The radar beacon shall have 1 delay setting which shall be 1.0 ±0.1 microseconds at a signal level of 0 dbm.

3.4.2.4.3 Delay variation with signal level. The total (absolute) variation in delay shall not exceed 0.2 microseconds for signal levels of 0 dbm to within 3 db of threshold.

3.4.2.4.4 Delay jitter. Delay jitter, which is defined as pulse to pulse variation in delay as a constant input signal level, shall not exceed the following values.

a. For signal levels of 0 dbm to -55 dbm: 0.035 microseconds peak-to-peak.

b. For signal levels of -55 dbm to threshold: 0.05 microseconds peak-to-peak.

3.4.2.5 Pulse recurrence frequency (PRF). The equipment shall be capable of operation at any PRF from 10 to 2600 pulses per second (pps), unless this would result in exceeding the duty cycle limits specified in 3.5.1.7, in which case the maximum PRF shall be that corresponding to the maximum allowable duty cycle. For RT-763D/APN-164(V) and RT-763E/APN-154(V), the equipment shall be capable of operation at any PRF from 10 to 4200 pps.

3.4.2.6 Recovery time. When operated in either the single pulse or coded interrogation mode, within the duty cycle limitations of 3.4.1.7, the radar beacon shall be capable of replying to each of 3 interrogations (code groups) when the spacing between any two successive interrogations is not less than 50 microseconds. These requirements shall be satisfied even though the signal level of adjacent interrogations differs by as much as 60 dB. The peak power of the reply to each interrogation shall be within 1 dB of the limits specified in 3.4.2.11. The radar beacon recovery time shall not be less than 13 microseconds.

3.4.2.7 Random triggering. The equipment shall not generate random replies (those replies not synchronized with interrogations) at a rate greater than 10 pps under any operating conditions specified herein.

3.4.2.8 Sensitivity and dynamic range. The radar beacon shall reply to at least 99 percent of the interrogation signals at any input signal level between 0 dbm and -65 dbm under standard conditions. Under extreme conditions, sensitivity shall be at least -62 dbm. In the coded mode of operation, the sensitivity shall be measured at the optimal code spacing, which shall be within ±0.1 microseconds of the nominal values defined in 3.5.1.3.2.

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3.4.2.9 Over interrogation protection. The radar beacon shall not suffer permanent damage when subjected to interrogation of PRFs which would result in exceeding the duty cycle limitation specified in 3.5.1.7. Within 5000 µs after the interrogation PRF falls to the maximum specified value or less, the radar beacon shall meet all requirements of this specification.

3.4.2.10 Impedance. The nominal input and output impedance of the radar beacon shall be 50 ohms.

3.4.2.11 Power output. The peak power output of the radar beacon shall be not less than 200 watts under all conditions specified herein. For RT-763E/APN-154(V), the peak power output of the radar beacon shall be not less than 300 watts under all conditions specified herein. Power output shall be measured at the antenna terminal of the duplexer-circulator.

3.4.2.12 Transmitter frequency stability. Under any operating conditions specified herein, except during changes in ambient temperature, the transmitter shall not drift more than ±5 MHz and the drift rate shall not exceed 1 MHz per minute after a five minute warm up. During changes in ambient temperature, the temperature coefficient of frequency drift of the transmitter shall not exceed 50 kHz per degree celsius. This shall include any frequency modulation present during service conditions as specified herein.

3.4.2.13 Transmitter short term stability. For RT-763E/APN-154(V), beginning ten seconds interrogation starts and during the next five minutes, transmitter frequency shall not change more than 5 MHz.

3.5 Detail requirements

3.5.1 Receiver-transmitter. There shall be twelve configurations as listed in table I. All configurations shall operate as described in the following paragraphs. The differences are in form factor, mounting, weight, turning range, and inclusion of an elapsed time indicator. The contract shall specify the desired configuration.

3.5.1.1 Receiver frequency stability. Under any operating conditions specified herein, except during changes in ambient temperature, the receiver frequency shall not vary more than ±3 MHz after a five minute warm up. During changes in ambient temperature, the coefficient of frequency drift shall not exceed 75 kHz per degree celsius. However, changes in receiver frequency due to ambient temperature changes shall not require a change in the interrogating signal frequency to meet the sensitivity requirements of 3.4.2.8.

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TABLE I. Receiver-transmitter configuration

Item	Form factor	Maximum weight (lbs)	Tuning range	Mounting	Elapsed time indicator
RT-681/APN-154(V)	Figure 1	4.0	9000-9600 MHz	Flange	No
RT-762/APN-154(V)	Figure 2	4.4	8500-9100 MHz	Side	No
RT-763/APN-154(V)	Figure 2	4.4	9000-9600 MHz	Side	No
RT-764/APN-154(V)	Figure 1	4.0	8500-9100 MHz	Flange	No
RT-681A/APN-154(V)	Figure 3	4.3	9000-9600 MHz	Flange	Yes
RT-762A/APN-154(V)	Figure 4	4.7	8500-9100 MHz	Side	Yes
RT-763A/APN-154(V)	Figure 4	4.7	9000-9600 MHz	Side	Yes
RT-764A/APN-154(V)	Figure 3	4.3	8500-9100 MHz	Flange	Yes
RT-763B/APN-154(V)	Figure 4	4.4	9000-9600 MHz	Side	Yes
RT-763C/APN-154(V)	Figure 4	4.4	9000-9600 MHz	Side	Yes
RT-763D/APN-154(V)	Figure 4	4.4	9000-9600 MHz	Side	Yes
RT-763E/APN-154(V)	Figure 4	4.4	9000-9600 MHz	Side	Yes

3.5.1.2 Selectivity

3.5.1.2.1 Bandwidth. The receiver bandwidth shall be not less than 8 MHz not more than 14 MHz as measured at the points corresponding to a 3 dB decrease from the maximum sensitivity.

3.5.1.2.2 Image. The image frequency shall be not less than 100 MHz from the nominal frequency. At the image frequencies, the transponder shall exhibit 40 dB decrease in sensitivity compared to that at the frequency to which the receiver is tuned.

3.5.1.3 Input signal. The receiver-transmitter shall be capable of operation from several types of input signals depending on the code chosen. A choice of single pulse operation or operation with any one of 5 double pulse codes shall be possible when used with the external (remote) control specified herein.

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3.5.1.3.1 Single pulse operation. In this mode of operation, the transmitter shall reply to a single pulse of the following characteristics:

a.	Pulse width	0.3 to 3 µs
b.	Rise time	0.1 µs or less
c.	Frequency	Within ±2 MHz of the frequency to which the receiver is tuned

3.5.1.3.2 Double pulse operation. The receiver-transmitter shall respond to any one of five spacings of double pulse interrogation as follows:

a. 1.9 ±0.1 µs - Mode 1

b. 2.8 ±0.1 µs - Mode 2

c. 3.7 ±0.1 µs - Mode 3

d. 4.6 ±0.1 µs - Mode 4

e. 5.5 ±0.1 µs - Mode 5

Individual pulse characteristics area:

a.	Pulse width	0.3 to 0.75 µs
b.	Pulse rise time	0.1 µs or less
c.	Pulse full time	0.2 µs or less

3.5.1.3.2.1 Double pulse rejection characteristics

a. When the radar beacon is set for acceptance of a double pulse interrogation whose spacing is 1.9 microseconds, interrogations having a pulse spacing less than 1.2 microseconds or greater than 2.6 microseconds shall not cause a beacon response to be generated.

b. When the radar beacon is set for acceptance of double pulse interrogation whose spacing is 2.8 microseconds, interrogations having a pulse spacing less than 2.1 microseconds or greater than 3.5 microseconds shall not cause a beacon response to be generated.

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c. When the radar beacon is set for acceptance of a double pulse interrogation whose spacing is 3.7 microseconds, interrogations having a pulse spacing less than 3.0 microseconds or greater than 4.4 microseconds shall not cause a beacon response to be generated.

d. When the radar beacon is set for acceptance of a double pulse interrogation whose spacing is 4.6 microseconds, interrogations having a pulse spacing less than 3.9 microseconds or greater than 5.3 microseconds shall not cause a beacon response to be generated.

e. When the radar beacon is set for acceptance of a double pulse interrogation whose spacing is 5.5 microseconds, interrogations having a pulse spacing less than 4.8 microseconds or greater than 6.2 microseconds shall not cause a beacon response to be generated.

3.5.1.4 Input impedance. The receiver terminal of the receiver-transmitter shall nave a nominal input impedance of 50 ohms.

3.5.1.5 Transmitter pulse characteristics. The transmitter output pulse shall have the following characteristics:

a. Pulse width: 0.6 ±0.1 μs

b. Pulse width Jitter: Not to exceed 0.01 μs peak-to-peak at the one-half power points.

c. Rise time: Not to exceed 0.1 μs from 10 percent to 90 percent amplitude points.

d. Full time: Not to exceed 0.2 μs from 90 percent to 10 percent amplitude points.

e. Spectrum: The rf bandwidth (in megacycles) of the transmitter pulse, measured at one-quarter power by means of an rf spectrometer, shall not exceed 3.0 megacycles divided by the pulse width in microseconds.

3.5.1.6 Output impedance. The transmitter terminal of the receiver-transmitter shall have a nominal output impedance of 50 ohms.

3.5.1.7 Transmitter duty cycle. The transmitter shall meet the requirements of this specification at any duty cycle up to 0.002.

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3.5.1.8 VSWR of load. The transmitter shall meet the requirements of this specification and shall not drift more than ±0.75 MHz when operating into the duplexer CU-1104/APN-154(V) with a load at the circulator antenna terminal having a mismatch such as to cause VSWR having a value varying up to 1.5 at any phase angle. In addition, the transmitter shall meet all of the requirements of this specification after application and removal of either a short circuit or an open circuit at the duplexer-circulator antenna terminal.

3.5.1.9 Function. The receiver-transmitter receives interrogations from X-band radars operating on the same frequency that the beacon receiver is tuned to; the interrogation signal is decoded and is used to trigger the beacon transmitter, which is tuned to another frequency within the X-band.

3.5.1.10 Contents. The receiver-transmitter shall include subassemblies that perform the following functions:

a. Preselector

b. Mixer

c. Local oscillator

d. IF amplifier

e. Decoder

f. Modulator

g. Power supply

h. Magnetron

3.5.1.11 Form factor. The form factor shall be as shown in Table I.

3.5.1.12 Weight. The weight of this unit shall not exceed that shown in Table I.

3.5.1.13 Controls. The following controls shall be provided on the external surface of the receiver-transmitter.

a. Preselector frequency control

b. Local oscillator frequency control

c. Transmitter frequency control

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Provisions shall be made to prevent inadvertent movement of these controls. Adjustment of those controls shall require a hand tool of some type. Pressurized covers shall be provided to maintain receiver-transmitter internal pressure and to protect controls when not in use.

3.5.1.14 Electrical connections. Connections to external circuits shall be provided as indicated in Table II.

Table II. Electrical connections

Reference designation	Receptacle type (Or an approved equivalent)	Function
J-1	Cannon Electric Company #MC-14H-10-6PN	Power and test
J-2	General RF Fittings, Inc. #182 (Female, Coaxial TNC)	Receiver connection
J-3	Coaxial TNC, Female (Part of magnetron)	Transmitter connection
J-4	Cannon Electric Company #MC-14H-10-6PN	Connect to control for code selection

3.5.1.15 Test points

3.5.1.15.1 External test point. A pulse signal equivalent to the decoder input signal and indicative of the transmitter reply shall be brought out to external connector for test use while the transponder is operating. The amplitude of this signal shall be at least one tenth of a volt peak into a 470 ohm load.

3.5.1.15.2 Internal test points. Sufficient test points shall be provided inside the transponder set to allow rapid isolation of malfunctions to a particular section of the transponder.

3.5.1.16 Pressurization. The receiver-transmitter shall be capable of maintaining an internal pressure of 30 psi above the external pressure. The following provisions shall be met:

a. The case shall be of a type that will permit ready access for alignment, testing, and repair. The receiver-transmitter shall be fully operable with the case opened, and the tuning adjustments of the receiver-transmitter shall not be affected by opening or closing the case.

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b. Means shall be provided for pressurizing the receiver-transmitter to 30 psig and measuring this pressure to determine the effectiveness of the seal. An air valve (not part of the radar beacon) shall be used for this purpose.

c. The seal shall be effective after repeated openings and resealing under field conditions.

3.5.1.17 Elapsed time indicator. An elapsed time indicator shall be provided for the configurations shown in Table I. The elapsed time indicator shall be attached to the R-T unit and shall measure the total time that 28 Vdc has been applied to the R-T unit. The indicator shall have a 1000-hour range.

3.5.2 Control C-4419/APN-154(V). The control shall meet the applicable requirements of MIL-C-6781 for type 1 control panels.

3.5.2.1 Function. This unit shall provide controls for tuning on the equipment and for selecting the mode of decoder operation.

3.5.2.2 Contents. This unit shall contain a toggle switch for turning equipment off and on, a rotary selector switch for selecting the mode of decoder operation and may contain passive circuit components.

3.5.2.3 Form factor. This unit shall have a maximum allowable size of 5.75 inches wide by 1.5 inches long by 2 inches deep.

3.5.2.4 Weight. The weight of this unit shall not exceed 0.75 pounds.

3.5.2.5 Controls. The following controls shall be provided:

a. "PWR" (Power). A three-position switch shall be provided for controlling the power to the equipment. The position shall be marked "off," "standby", and "operate".

b. MODE. A rotary selector switch shall be provided for selecting the mode of operation of the equipment. Position shall be as follows:

1. Single

2. Double - 1

3. Double - 2

4. Double - 3

5. Double - 4

6. Double - 5

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3.5.2.6 Electrical connections. Connections to external circuits shall be provided by means of a standard connector in accordance with MIL-STD-5400.

3.5.3 Duplexer CU-1104/APN-154(V)

3.5.3.1 Function. The duplexer shall perform the following functions:

a. Connect the receiver and transmitter antenna terminals of the receiver-transmitter in such a manner that the radar beacon may operate from a single antenna for both transmitting and receiving. The transmitter and receiver frequency shall be separated by at least 60 MHz.

b. Isolate the rf output of the transmitter from the "long line" effects of the antenna, which might detune the transmitter an excessive amount.

c. Isolate the receiver input from the transmitter output to prevent damage to the mixer crystal.

3.5.3.2 Form factor. The duplexer shall not exceed 2.16 inches by 1.22 inches in cross section (excluding connectors) and 0.89 inch in thickness.

3.5.3.3 Weight. The weight of the duplexer shall not exceed 0.5 pound.

3.5.3.4 Electrical connections. The external connectors shall be coaxial TNC.

3.5.4 Cable assembly, radio frequency CG-2574/APN-154(V) and CG-2574A/APN-154(V)

3.5.4.1 Function. Two interconnecting cables shall be provided to connect the receiver-transmitter to the duplexer.

3.5.4.2 Form factor. The cable shall not exceed 6 inches in length and shall be adequate to allow the equipment to meet the requirements specified herein. For CG-2574A/APN-154(V), the cable length shall not exceed 8 inches.

3.5.4.3 Weight. The weight of each cable shall not exceed 0.1 pounds.

3.5.5 Control C-4419/APN-154(V). The control shall be the same as C-4419/APN-154(V) (see 3.4.2), except an elapsed time indicator shall be provided. The form factor shall not change.

3.5.5.1 Weight. The weight of this unit shall not exceed 0.85 pound.

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3.5.5.2 Elapsed time indicator. The indicator shall be attached to the front panel and shall measure the total time 28 Vdc has been applied to the control. The indicator shall have a 1000 hour range.

3.5.6 Control C-4419B/APN-154(V). The control shall meet the applicable requirements of MIL-C-6781 for type 1 control panels.

3.5.6.1 Function. This unit shall provide controls for turning on the radar beacon systems, for selecting the mode of decoder operation, and for selecting and testing the ACLS mode of operation when operated with the associated equipment listed in 6.11.

3.5.6.2 Form factor. The control shall have a maximum allowed size of 5.75 inches width 1.5 inches height and a 2 inch depth, excluding the integral interconnecting cable and front panel controls. Provisions shall be made for mounting on the aircraft cockpit console.

3.5.6.3 Controls. The following controls shall be provided:

a. POWER. A three position switch shall be provided for controlling power to the radar beacon system and, when activated, the ACLS system. The positions shall be labeled "OFF," "STBY", and "PWR". The "PWR" positions shall enable input primary power to be applied to the radar beacon system. The ACLS position shall enable primary input power to be applied to both the radar beacon system and the augmenter/ receiver R-1623/APN. The "STBY" position shall disable the decoder assembly. The "OFF" position shall disconnect primary input power.

b. MODE. A rotary selector switch shall be provided for selecting the mode of operation of the system. The switch, when in ACLS position, shall also provide for processing Ka-band interrogations signals in augmenter/receiver R-1623/APN, which provides external trigger for the RT-763/APN-154(V). Positions shall be as follows:

ACLS, SINGLE, DOUBLE-1, DOUBLE-2, DOUBLE-3, DOUBLE-4, DOUBLE-5

c. ACLS TEST. A push-to-test switch with a dimming type indicator light shall be provided. When the MODE switch is in ACLS position, the ACLS TEST switch shall provide for illumination during normal ACLS receiver operation if interrogated by Ka-band signals. When processed, the switch shall provide for a test signal (BIT) for testing the augmenter/receiver when Ka-band interrogations are not present.

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3.5.6.4 Panel. The front panel of the control shall be integrally illuminated in accordance with MIL-P-7788, class 1-R, type IV. Unless otherwise specified in the contract, illumination shall be operated from a 5 volt aircraft panel lighting circuit. Unless otherwise specified in the contract, the lighting shall be red.

3.5.6.5 Electrical connections. Connections to external circuits shall be provided by means of an integral cable terminated in a standard connector in accordance with MIL-STD-5400.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Production sample tests (see 4.4)

c. Quality conformance inspection (see 4.5)

d. Life test (see 4.6)

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* 4.3 First article inspection. First article inspections shall be made on equipment representative of the production equipment to be supplied under the contract (see 6.5). First article inspections shall consist of the following tests:

a. Contractor's demonstration tests (see 4.3.1)

b. Service approval tests (see 4.3.2)

4.3.1 Contractor's demonstration tests. Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.7. The Government quality assurance representative and the acquiring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.3.1.1 Demonstration test data. The contractor shall submit all data collected in conducting these tests to the acquiring activity for review and approval. On production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and ambient surface temperatures. The ambient and surface temperatures shall be obtained under the extreme high temperatures operating condition. Such data on parts may be included as part of the preliminary parts list.

4.3.2 Service approval tests. At the completion of the contractor's demonstration tests and when requested by the acquiring activity, the equipment shall be delivered to a specified Government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.3.2.1 Accesory material. In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design test data specified in MIL-STD-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

* 4.3.3 Scope of tests. First article inspections shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. First article inspection shall include environmental tests in accordance with MIL-STD-810 and interference tests in accordance with MIL-STD-462.

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* 4.3.4 First article approval. The acquiring activity shall approve the first article sample upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the first article sample. Prefabrication of production equipment prior to the approval of the first article sample is at the contractor's own risk. The approved first article sample will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for quality conformance. The first article sample shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3.5 Production equipments. Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance, and quality, be equivalent to the approved first article sample, Each equipment shall be capable of successfully passing the same tests imposed on the first article sample. Evidence of non-compliance with these requirements shall constitute cause for rejection. For equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the acquiring activity.

4.4 Production sample tests. When requested by the acquiring activity, one equipment shall be selected from the first ten production equipments submitted for acceptance and sent to a designated Government laboratory for tests. The equipment shall be selected by the Government quality assurance representative after the equipment has successfully passed all individual tests.

4.4.1 Scope of tests. The equipment may be subjected to any and all tests the acquiring activity deems necessary to ensure the production equipment is equivalent to the previously approved first article sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements.

4.4.2 Production sample approval. Upon satisfactory completion of all tests, approval of the production sample shall be by the acquiring activity. Any design, material, or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the acquiring activity. Failure of the production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the acquiring activity.

4.4.3 Reconditioning of production test sample. On completion of the production sample test, the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

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* 4.5 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance tests. All inspection and testing shall be under the supervision of the government quality assurance representative. Contractors not having testing facilities satisfactory to the acquiring activity shall engage the service of a commercial laboratory acceptable to the acquiring activity. The contractor shall furnish test reports showing quantative results for all quality conformance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance tests consist of the following:

a. Individual tests (see 4.5.1)

b. Sampling tests (see 4.5.2)

c. Reliability assurance tests (see 4.5.3)

d. Special tests (see 4.5.4)

4.5.1 Individual tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of product (see 4.5.1.1)

b. Operational test (see 4.5.1.2)

c. Manufacturer's run-in test (see 4.5.1.3)

4.5.1.1 Examination of product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.5.1.2 Operational test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to ensure satisfactory equipment operation.

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4.5.1.3 Manufacturer's run-in test. NOTE: This test shall not be required if a reliability test is selected that requires a burn-in test for all equipments. Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ±g's.

The equipment shall be vibrated (without vibration insulators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions, the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours. (NOTE: This test shall not be required if a reliability test is selected that requires a burn-in test for all equipments.)

4.5.2 Sampling tests. Sampling tests shall be conducted on equipment which have successfully passed the individual tests. Equipments shall be selected for sampling tests by the Government quality assurance representative in accordance with the following. (When reliability assurance tests of 4.5.3 are conducted, sampling tests shall be omitted.)

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.5.2.1 Scope of tests. As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

a. Complete operational test at ambient room conditions, making all necessary measurements to ensure all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the first article inspection, production and special tests.

c. Manufacturing reliability test specified in 4.5.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failures shall be analyzed as to the cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.5.3 Reliability assurance tests. Reliability assurance tests shall be conducted in accordance with MIL-STD-781. Tests as required by both the qualification phase and the sampling phase.

4.5.3.1 Qualification phase. Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests." The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the qualification phase, test level E shall be used. The accept-reject criteria for test plan II shall be used.

4.5.3.2 Reliability sampling phase tests. The reliability sampling phase test shall be accomplished on 100 percent of all production equipment not subjected to the reliability qualification phase test. Each equipment produced shall be subjected to a "burn-in" of two complete cycles as defined

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in 4.5.3.3. This burn-in may be extended at the option of the contractor. Following the burn-in, the equipment shall be tested for eighteen operating hours (four complete cycles of 4.5.3.3).

To determine whether the MTBF is being met at any time during the contract, the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totalled and the results compared with the reject line of test plan II of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that any one time the experience from the beginning of the contract is included. At the conclusion of each month, the test results shall be sent to the acquiring activity and to the Naval Air Systems Command, Attn: Avionics Division. At any time that the current totals of test hours and test failures plotted on test plan II curves show a reject situation, the acquiring activity shall be notified. The acquiring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a rejection decision.

4.5.3.3 Duty cycle. The duty cycle to be used shall be in accordance with test level E of MIL-STD-781, and shall consist of a 4-1/2 hour heating period (equipment on) followed by a 1-1/2 hour cooling period (equipment off).

4.5.3.4 Test details. The test details such as the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the acquiring activity prior to the beginning of the qualification test phase of the reliability assurance tests.

4.5.4 Special tests. Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to ensure adequate quality control. The equipment selected for special tests may be selected from equipment previously subjected to the sampling tests.

4.5.4.1 Special test schedule. Selection of equipment for special tests shall be made as follows, except that when a production sample is requested and delivered to a Government laboratory for test (see 4.4), the requirement for a special test on an early production equipment may be omitted.

a. On an early production equipment.

b. On an early equipment after an engineering change.

c. Whenever failure reports or other information indicate additional equipment should be tested. (This will be determined by the acquiring activity.)

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4.5.4.2 Scope of tests. Special tests shall consist of such tests as are approved by the acquiring activity. Test procedures previously approved for first article and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the acquiring activity for approval prior to conducting the tests. As a minimum, the following test shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

a. Temperature - Altitude tests.

b. Vibration test (60 minutes cycling in each of 3 planes).

c. Humidity tests.

4.5.5 Equipment failure. Should a failure occur during either the sampling or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or design defect.

c. Submit to the acquiring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

d. Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.6 Life test. The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.6.1. The life test sample shall be selected by the Government quality assurance representative in accordance with the following, except that when a production sample equipment is requested and submitted to the acquiring activity for test then the first equipment specified below need not be selected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.6.1 Test conditions. The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room.
Altitude	Normal ground (0-5000 ft)
Humidity	Ambient room
AC Voltage	115 ±5 volts (at lowest applicable frequency).
DC Voltage	27.0 ±2.00 volts

4.6.2 Test periods. The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.6.3 Performance check. At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the acquiring activity.

4.6.4 Test data. The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

a. Part type number.

b. The circuit reference symbol number.

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c. The part function.

d. Name of the manufacturer.

e. Nature of the failure.

f. The number of hours which the part operated before failure.

4.6.4.1 Failure report. In the event of a failure, the Government quality assurance representative shall be notified immediately. A report shall be submitted to the acquiring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design of material corrections for all failures which occurred. The acquiring activity will review such proposals and determine whether they are acceptable.

4.6.5 Reconditioning of life test samples. An equipment which has been subjected to the life test shall be reconditioned as follows:

a. On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the acquiring activity.

b. After reworking, the contractor may resubmit the equipment for acceptance.

4.7 Test procedures. The procedures used for conducting first article inspection, acceptance tests and life tests shall be prepared by the contractor and submitted to the acquiring activity for review and approval. The right is reserved by the acquiring activity or the Government quality assurance representative to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the acquiring activity.

4.8 Presubmission testing. No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge, with all applicable requirements.

4.9 Rejection and retest. Equipment that has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the Government quality assurance representative.

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5. PACKAGING

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or purchase order in accordance with MIL-E-17555.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The radar beacon shall be for use in Naval aircraft in conjunction with radar direction finding and ranging to increase the range capability of the system.

6.2 Test values. Normal and limiting values of performance data shall be determined at input voltages of 28 volts ±1.0 Vdc. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

* 6.3 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, including any amendments.

b Type designation and quantity of equipment required (see 1.2, 1.3, 1.5, and 1.6).

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

d. When first article is require (see 3.1).

e. Applicable levels of preservation, packaging, and packing (see 5.1).

* 6.4 Data requirements. The following Data Item Descriptions (DID's) must be listed, as applicable, on the Contract Data Requirements List (DD Form 1423) when this specification is applied on a contract, in order to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.3.1.1	DI-NDTI-80809A	Test/inspection reports	---

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4.5.3.4	DI-NDTI-80603	Test procedure	---
4.6.4.1	DI-R-21598	Failure reports	---

The above DID's were those cleared as of the data of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure only current, cleared DID's are cited on the DD Form 1423.

* 6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.3.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply.

6.7 Type designation. The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the acquiring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.8 Performance objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. When it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance, or reliability will result from the use of materials, parts and processes other than those specified in MIL-STD-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the acquiring activity for consideration. Each request shall be accompanied by complete supporting information.

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6.9 Definitions. The following definitions shall apply:

6.9.1 Interrogation. An interrogation is one or two radio frequency pulses transmitted by an interrogator or radar set.

6.9.2 Pulse amplitude. Pulse amplitude is the maximum voltage amplitude of the pulse or rf pulse envelope, excluding spike.

6.9.3 Pulse fall time. Pulse fall time is that interval of time required for the trailing edge of the pulse to decay from 90 to 10 percent of the pulse amplitude.

6.9.4 Pulse rise time. Pulse rise time is that interval of time required for the leading edge of the pulse to rise from 10 to 90 percent of the pulse amplitude.

6.9.5 Pulse spacing. Pulse spacing is that interval of time between 50 percent pulse amplitude points on the leading edges of the pulses involved.

6.9.6 Pulse width. Pulse width is that interval of time between points at which amplitude of the pulse is 50 percent of the pulse amplitude.

6.9.7 Reply. A reply is a radio frequency pulse transmitted by a beacon as a result of an interrogation.

6.9.8 Threshold signal level. The minimum signal level, measured at the radar beacon antenna terminal that causes the radar beacon to reply to at least 99 percent of the interrogations per second.

6.10 Nonrepairable subassemblies. As a general rule, nonrepairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be minimal. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the nonrepairable subassembly should evidence a mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.11 Configuration of units. The contract or purchase order is to specify the configuration of the receiver-transmitter unit, the control, the duplexer, and the cable assembly.

6.12 Associated equipment. The radar beacon should be capable of operating as part of the ACLS in conjunction with the following equipment:

a.	Modulator, beacon	MD-817/APN
b.	Receiver, radar	R-1623/APN

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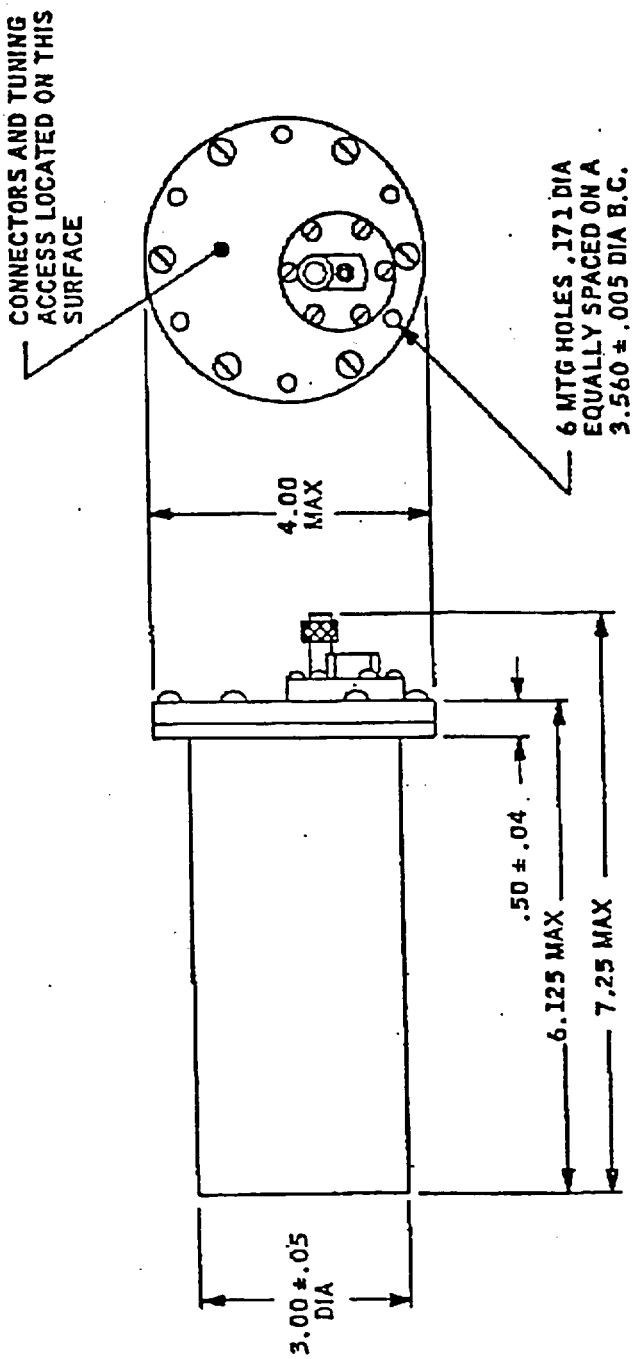
* 6.13 Subject term (keyword) listing.

Band
Cable assembly
Control
Duplexer
Modulator
Pulse
Receiver
Signal transmitter

* 6.14 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the past previous issue.

Preparing activity:
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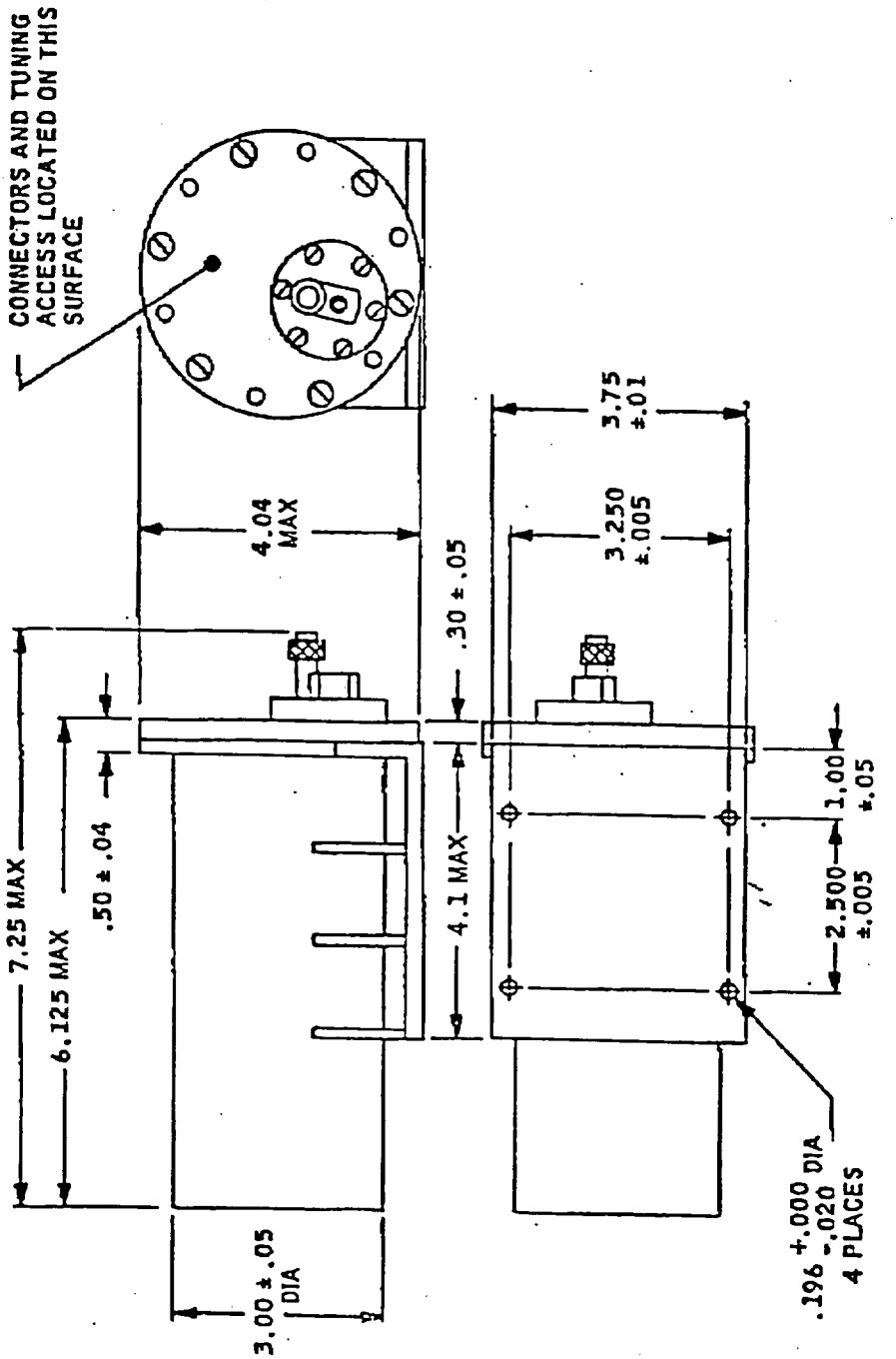


Type designation:

RT-681/APN-154(V) High band receiver
RT-764/APN-154(V) Low band receiver

FIGURE 1. Receiver-transmitter, flange mount, design control drawing.

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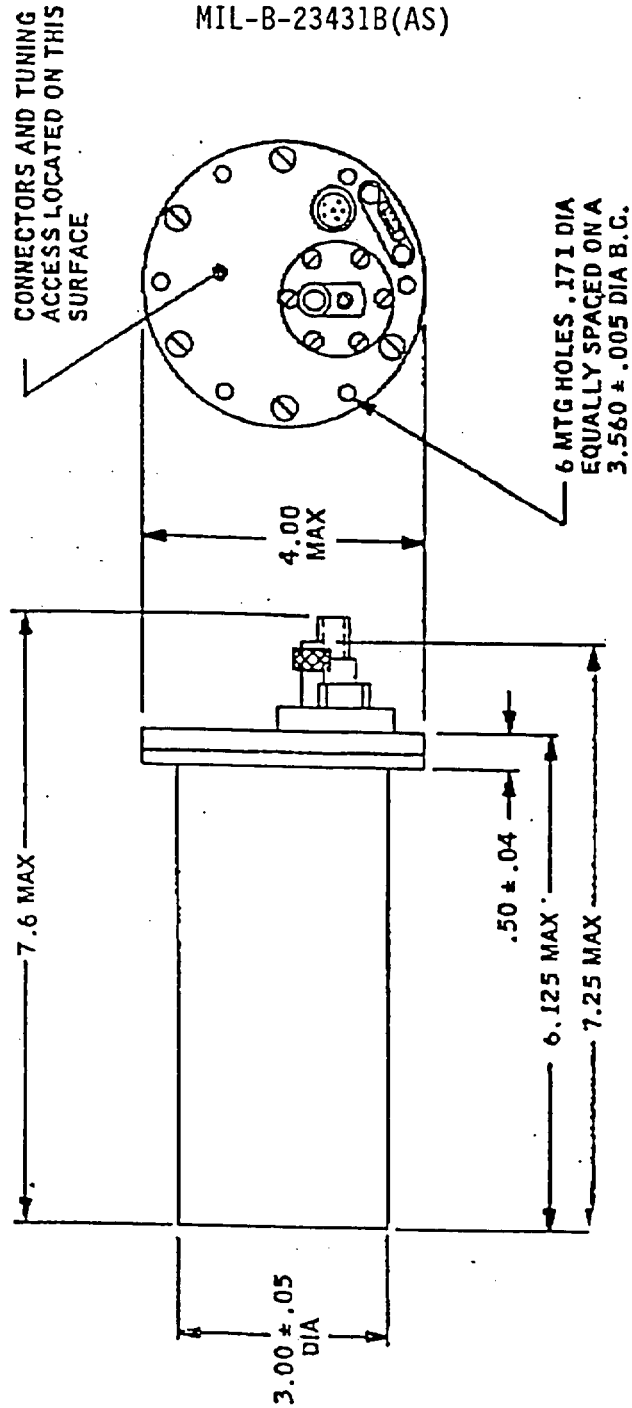


Type designation:

RT-762/APN-154(V) Low band receiver
RT-763/APN-154(V) High band receiver

FIGURE 2. Receiver-tranmsitter, side mount, design control drawing.

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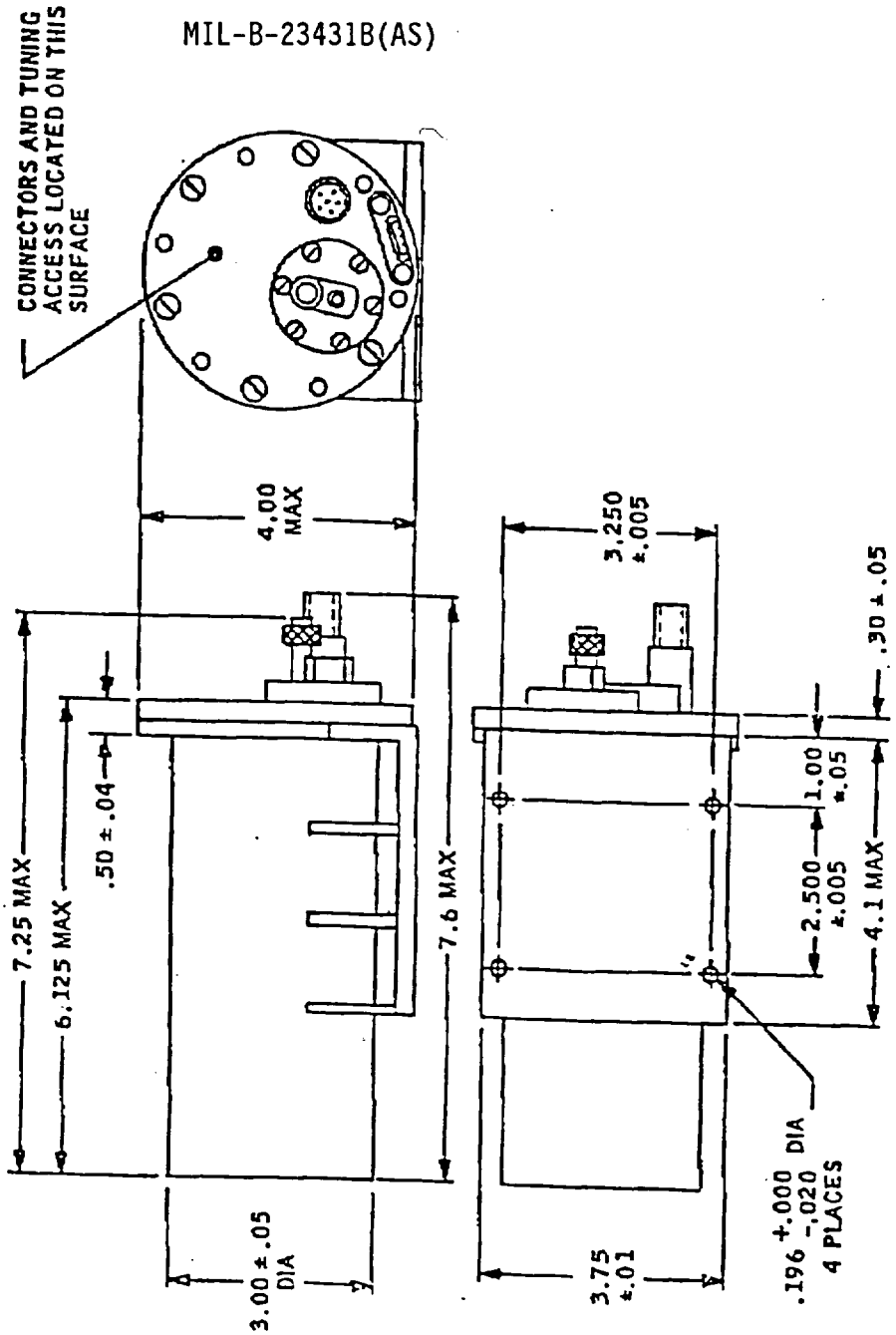


Type designation:

RT-681A/APN-154(V) High band receiver
RT-764A/APN-154(V) Low band receiver

FIGURE 3. Receiver-transmitter, flange mount with elapsed time indicator, design control drawing.

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Type designation:

RT-762A/APN-154(V) Low band receiver
RT-763A/APN-154(V) High band receiver
RT-763B/APN-154(V) High band receiver
RT-763C/APN-154(V) High band receiver
RT-763D/APN-154(V) High band receiver
RT-763E/APN-154(V) High band receiver

FIGURE 4. Receiver-transmitter, side mount with elapsed time indicator, design control drawing.

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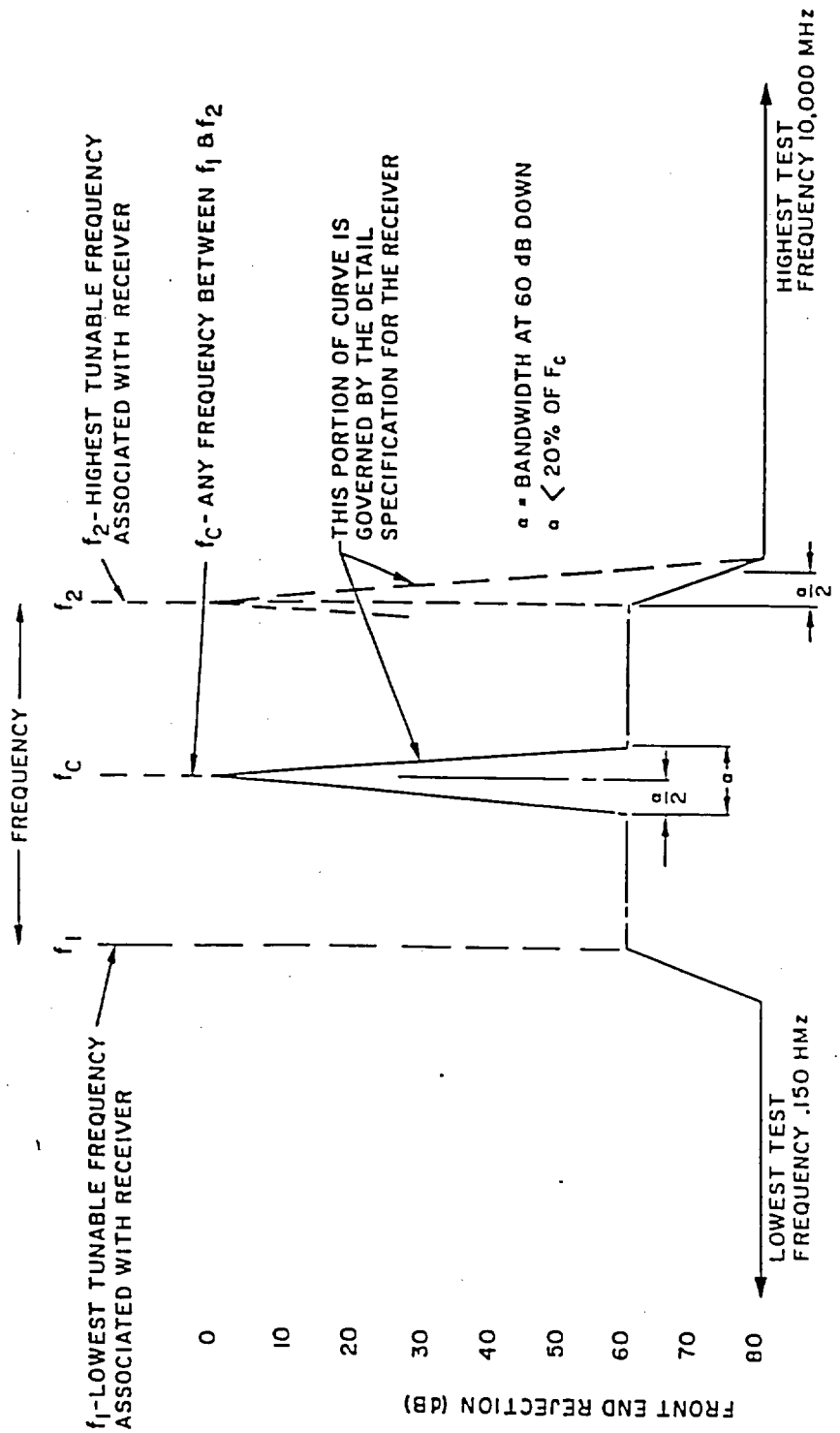


FIGURE 5. Require receiver front-end rejection.

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